



NIRMAL KUMAR BV

QA AUTOMATION ARCHITECT & CONSULTANT

S U M M A R Y:

Quality Engineering Architect and Leader with close to 15 years of experience driving enterprise-scale test strategy, automation transformation, and engineering excellence across web, mobile, and API platforms. Enthusiastic about AI-driven automation, recognized for building high-performing teams, leading cross-functional delivery, and enabling predictable releases through scalable, intelligent, and data-driven quality practices.

E X P E R I E N C E:

QA AUTOMATION ARCHITECT

KNOW, CHENNAI

2024-2026

KNOW is a unified mobile platform transforming frontline operations by digitizing workflows, training, and workforce management across industries, including logistics, construction, retail, and hospitality.

- Architected a scalable, unified automation framework for web and mobile using Selenium, Appium, and TestNG, with extensibility for API testing using Rest Assured.
- Led an end-to-end automation strategy, including framework design, tool selection, coding standards, and best practices, to improve test reliability and maintainability.
- Integrated automation with Lambda Test cloud for parallel cross-browser and cross-device execution, improving execution efficiency and coverage.
- Automated critical sanity and regression suites across key business modules to improve release confidence and reduce manual effort. Designed and implemented CI/CD pipelines in Bitbucket, enabling automated quality gates via manual triggers and scheduled runs.
- Mentored QA engineers and enabled team adoption of automation through training and knowledge sharing. Collaborated with cross-functional stakeholders to align the quality strategy with the product roadmap and release cycles.
- Maintained strong communication with clients throughout the design process to ensure alignment between their vision and the final product.
- Improved overall automation maturity by establishing reusable components, modular design patterns, and scalable execution strategies.



QA AUTOMATION ARCHITECT

PICKYOURTRAIL, CHENNAI

2023-2024

Pickyourtrail is a travel-tech startup delivering customized international travel experiences through a digital platform, enabling customers to explore destinations at their own pace.

- Led an end-to-end Quality Engineering strategy across Web, Mobile, API, and performance testing, driving scalable automation adoption across the organization.
- Scaled automation frameworks using Selenium, Playwright, and Appium, achieving approximately 75% coverage across core features, and improving release confidence.
- Introduced modern tooling, including Playwright and Cypress, to automate critical user journeys, improving stability and feedback cycles.
- Designed and implemented API automation using Rest Assured, validating approximately 75 critical APIs that support key B2B workflows.
- Led performance testing initiatives using JMeter to evaluate system scalability and identify bottlenecks.
- Established automation standards, reusable components, and best practices improve maintainability and scalability.
- Mentored QA engineers, and enabled broader automation adoption across teams.
- Contributed to architecture discussions and technical decision-making, aligning quality strategy with delivery timelines.
- Collaborated with product and engineering stakeholders to define the test strategy and ensure release readiness.

QUALITY ASSURANCE MANAGER/TEST AUTOMATION TEAM LEAD

PICKYOURTRAIL, CHENNAI

2019-2023

- Led the Quality Engineering function, driving automation strategy and delivery excellence across Web, Mobile, API, and performance testing.
- Partnered with founders and leadership to align quality strategy with the product roadmap and business goals, ensuring predictable releases.
- Defined and tracked engineering KPIs, driving continuous improvement in quality, efficiency, and release readiness.



- Built and scaled high-performing QA teams, fostering ownership, accountability, and a strong engineering culture.
- Led architecture discussions and designed scalable automation frameworks using Selenium, Cucumber, Appium, and Rest Assured.
- Implemented parallel execution infrastructure using Zalenium and led migration to Kubernetes for scalable, containerized test execution.
- Established QA governance, quality gates, and best practices, improving engineering discipline and consistency.
- Collaborated with cross-functional teams in planning, design reviews, and release readiness decisions.
- Led hiring and capability-building initiatives, strengthening team technical depth.
- Delivered quality transformation during a high growth phase, enabling faster releases and improved product stability.
- Investigated production issues, and performed root cause analysis to drive systemic improvements.

SPECIALIST

CSS CORP PVT. LTD, CHENNAI

2015-2019

Blackboard is a leading Learning Management System (LMS), providing course delivery, content management, collaboration tools, and student assessment analytics for educational institutions.

- Automated approximately 75% of the regression test suite using Selenium and Java, improving test coverage and reducing manual effort.
- Integrated UI automation with REST APIs to enable end-to-end workflow validation across application layers.
- Developed 20+ automation and scripting utilities supporting Support, Operations, and DBA teams, improving efficiency, and reducing manual intervention.
- Automated routine operational tasks through custom tools, improving productivity, and turnaround time.
- Supported UAT and production releases, troubleshooting issues, and ensuring system stability.
- Collaborated with cross-functional teams to identify automation opportunities and improve testing efficiency.



- Contributed to improved release quality through early defect detection and continuous testing practices.

SENIOR DEVICE ASSOCIATE

AMAZON DEVELOPMENT CENTRE (INDIA) PVT LTD, CHENNAI

2010-2015

- Automated build and device flashing using Perl improves operational efficiency.
- Served as QA liaison across global teams for Kindle platforms.
- Conducted device testing, ensuring performance and release of readiness.
- Presented quality insights to leadership, and supported release decisions.
- Improved product reliability through defect identification and troubleshooting.
- Trained and onboarded new team members, improving team productivity and ramp-up time.

EDUCATION:

BACHELOR OF TECHNOLOGY: INFORMATION TECHNOLOGY

Anna University Anand Institute of Higher Technology - Rajiv Gandhi Salai, Kazhipattur

2005-2009

SKILLS/TOOLS:

QA Automation Architect & Consultant | Web, Mobile, API Testing | Exploring Self-Healing Automation Frameworks | Exploring AI-Based Test Generation & Optimization | Agentic QA Systems | Selenium | Playwright | Appium | Rest Assured | K6 | Core Java | Python | JavaScript | Typescript | Automation Frameworks (BDD, Hybrid) | Mentoring in Test Automation | Scraping Websites using Selenium and Python | Selenium Grid using Docker Selenium Grid using Kubernetes | AWS | Prompt Engineering | LLM-Assisted Testing | Test Strategy | Quality Metrics | Test Planning | Sprint & Release Planning | Risk Management | Hiring | Leaderships | Cypress | JMeter | Kubernetes | Python | GitHub Copilot